Asymmetric Capital Buffer Control

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1 Introduction

1.1 Question

Pending: the paper's question, in the author's words.

1.2 Answer

Pending: the answer, written from the model rows once they enter.

Table 1: Headlines.

ID	Headline	Model rows	Scope note

2 Model

2.1 Players

Table 2: Players of the model, the choice each faces and how the rows settle it.

ID	Player	Choice faced	How the model settles it	In the literature	Rows

2.2 Assumptions

Table 3: Assumptions of the model.

ID	Assumption	In plain English	Symbols	Model rows	Reason

2.3 Results

Table 4: Results of the model, each with its statement and the Lean theorems that prove it.

ID	Claim	Statement	Lean	Proof

3 Literature

Table 5: Papers read, with the passage each row relies on.

ID	Paper	What we rely on	Quote	Page	Rows

No paper is cited that was not read.

4 Conclusions

Table 6 holds the conclusions that rest on model rows and on papers read in full. Table 7 holds every claim that a result is new and every other claim still waiting, and each of its rows names what it waits on.

Table 6: Conclusions resting on model rows and papers read.

ID	Statement	Rows

Table 7: Claims not yet verified, each naming what it waits on.

ID	Statement	Rows

A Proofs

Notation. No symbol is defined yet. Each symbol is defined here once, when the first row that uses it enters.

B Evidence and verification

Every row of the model table will be a theorem of Lean 4 with Mathlib. The Lean column names the theorems, and the proof of each row in Appendix A names the same ones. A theorem cited as `Namespace.name` is declared in the file `theory/lean/Namespace.lean`. The development is built on the toolchain `v4.32.0-rc1` against Mathlib commit `360da6f`. The script `theory/verify.sh` runs the suites below on every build, and `theory/VERIFICATION.md` records what they printed.

1. *The Lean build and axiom audit.* Every file under `theory/lean/` is a build root and builds. No source contains `sorry`. Every declared theorem is audited with `#print axioms`, and none depends on an axiom outside `propext`, `Classical.choice` and `Quot.sound`. Each file is identical to its source in `lean/mathlib/`.
2. *The skeleton checker.* Rules MS-1 to MS-9 of `theory/checks/verify_manuscript.py`, each also run on a well-formed sample it must accept and a faulty sample it must reject.

What Lean does not carry. No model row has entered, so no assumption enters a theorem only as a hypothesis yet.

Illustrations. No row rests on SymPy or on numerical sampling.

C Terminology

Table 8: Terms, each bound to a symbol defined in Appendix A.

Term	Symbol	Meaning

D Primitives and scope

No primitive is stated yet. Each primitive and each boundary of the model gets one paragraph here, saying what lies outside the model and which rows depend on it.

E Refuted conjectures

No conjecture is refuted yet.

F Responses to the referee reports

Not applicable. The paper has not been through review.

G Coverage of the referee comments

Not applicable, for the same reason.